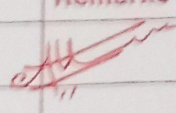
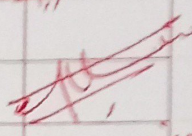
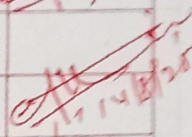
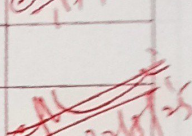
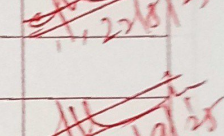
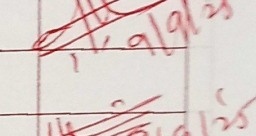
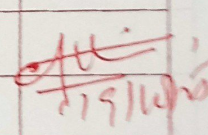
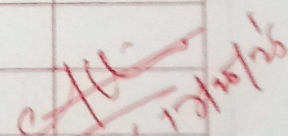
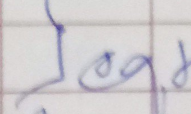
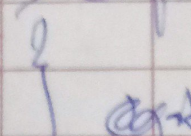


I N D E X

NAME: Kuberan STD.: A1 SEC.: A ROLL NO.: _____ SUB.: Deep learning

S. No.	Date	Title	Page No.	Teacher's Sign / Remarks
1	24/7	Exploring Deep learning Platform		
2	31/07	Implement a classifier using open source dataset		
3	14/08	study of the classifiers with respect to statistical parameters		
4	14/08	Build a simple feed forward neural network to register vc		
5	22/08	study of activation function and its role		
6	09/09	Gradient descent & backpropagation in deep neural networks		
7	16/09	Build a cnn model classify cat and dog image		
8	9/10	experiment of LSTM		
9	09/10	Build a Recurrent Neural Network		
10	17/10	Perform Compression on mnist dataset using Autoencoder		
11	17/10	Experiments using variational Autoencoder (VAE)		
12	26/10	Implement a Deep convolutional GAN (DCGAN) to color image		
13	26/10	Understand the Architecture of a pre-trained Model		
14	26/10	Implement a pre-trained CNN model as a Feature Extractor		
15	26/10	Implement a yolo model for object Detection		

26/10/25

Ex: No: 14

14. Implement a Pre-trained CNN Model as a Feature Extractor

Aim:

To use a pre-trained CNN model (e.g. ResNet 50, VGG16) as a feature extractor for a custom dataset.

Objectives:

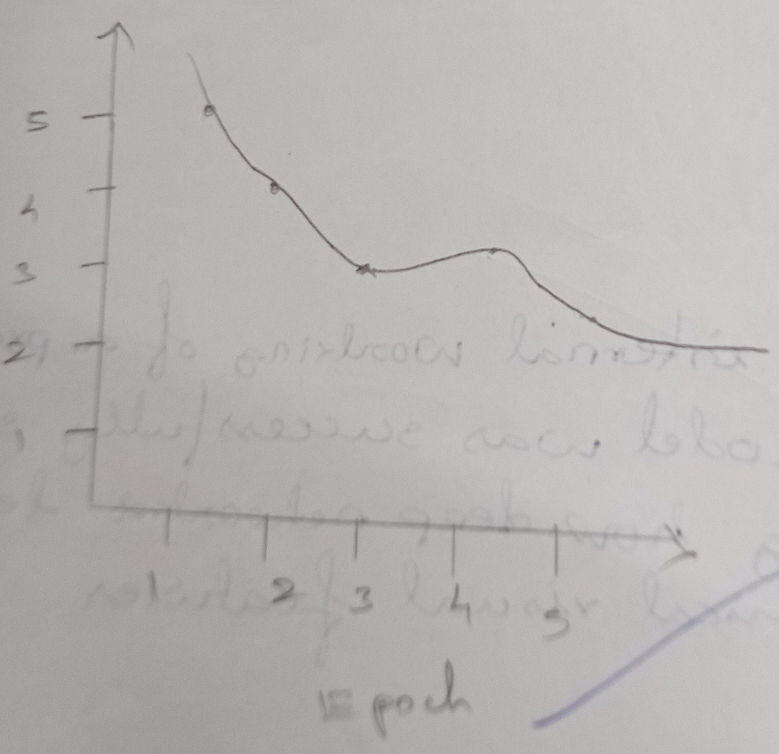
1. To understand Transfer learning
2. To reuse learned convolutional filters from large datasets
3. To train only the final classification layer with minimal computation

Pseudocode

1. start
2. Load pre-trained CNN model with weights = 'imagenet' and exclude top layers,
3. Freeze all convolution layers
4. compile model (Adam + Cross Entropy)
5. Train only the new top layers on custom dataset

and repeated training
 of the network
 layers and their
 representations
 A. The network model
 training time and inference
 accuracy

Graph



Result

The network model was
 trained for 5 epochs
 and the accuracy was
 2.5% at epoch 5.

Output

Epoch Loss

0.5

0.4

3

0.3

4

0.3

5

0.2

Test Accuracy : 88.4%

Sample Prediction

Input : Airplane

Output : Airplane

6. Evaluate model accuracy

7. END

Observation

1. Training was significantly faster compared to training from Scratch

2. Accuracy improved due to effective feature reuse

3. Freezing the base layers prevented ~~overfitting~~ on small datasets

Result

89. ✓ Transfer learning using a pre-trained CNN model successfully extracted features and improved classification performance efficiently